BINESH SADANANDAN

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Senior Principal R&D Engineer & Technical Fellow · Data Science, Machine Learning & GenAI

SUMMARY

Technical Fellow and Technical Lead for Artificial Intelligence (AI) Enablement in Medtronic's Surgical Operating Unit. Senior data science leader with 10+ years building production machine learning, generative AI, and enterprise data platforms in regulated healthcare, on a 19-year engineering foundation. Completed my Ph.D. in October 2026 at the University of New Haven, with research in medical AI safety and interpretability. Record includes 11 publications, two patents, and peer review service for ICLR, MICCAI, MLHC, and IEEE ICMLA.

CORE SKILLS

Machine Learning & GenAI: production ML (batch and real-time inference), predictive modeling, deep learning, LLM/VLM evaluation, retrieval-augmented generation (RAG), vector databases, prompt engineering

Responsible AI: robustness and bias evaluation, failure-mode analysis, model monitoring, clinical AI safety

Data Platforms: AWS, Snowflake, Redshift, Databricks, Azure ML, Dataiku, lakehouse and pipeline architecture, data governance, metadata management, engineering data management (PLM/Windchill), APIs and data services

Analytics & Visualization: Python, SQL, R, statistics, exploratory data analysis, anomaly detection, Power BI, Tableau, Plotly/Dash

Health Data: DICOM, MedDRA, clinical evidence generation

PROFESSIONAL EXPERIENCE

Medtronic – Surgical Innovation — **North Haven, CT · 2015 – Present**

Senior Principal R&D Engineer & Technical Fellow — 2023 – Present

- Serve as Technical Lead, AI Enablement, for the Surgical Operating Unit and as company-wide Technical Fellow, advising senior leadership on machine learning strategy and mentoring engineering teams across the organization.
- Lead data science and ML integration for surgical instruments, delivering predictive models and generative AI solutions for product development and device safety (AWS, Snowflake).
- Build retrieval-augmented generation (RAG) pipelines with vector databases for R&D document search and knowledge retrieval.
- Developed multimodal deep learning models for early prediction of ICU patient deterioration (Journal of Data Science, 2025) and generative models for early detection of device failures (ICMHI, 2024).
- Analyze MedDRA-coded adverse event data to support post-market device safety surveillance.
- Lead PLM and engineering data management practices for data integrity, traceability, and regulatory compliance.

Senior Principal R&D Engineer — 2021 – 2023

- Led scientific and clinical evidence strategies for minimally invasive surgical staplers, contributing to peer-reviewed publications and engagement with key opinion leaders.
- Designed and built scalable data pipelines with Dataiku, Python, Redshift, and Snowflake for clinical evidence generation and R&D decision support, with Power BI dashboards for stakeholders.
- Mentored junior data scientists and advised management teams on data-driven strategy and experimentation methods.

Principal R&D Applications Engineer — 2016 – 2021

- Drove new product development process improvements using cloud services, high-throughput event processing, and data analytics.
- Architected Windchill PLM solutions and business processes for R&D, translating business requirements into system design in an Agile environment.

Enterprise Solutions Consultant — 2015 – 2016

- Supported the R&D Model-Based Enterprise initiative with PTC Windchill and Integrity solutions.

EARLIER CAREER

Technical Architect (PLM) & Application Consultant – Barry-Wehmiller Design Group	2012 – 2015
Product Specialist & Enterprise Support Engineer – PTC	2010 – 2012
Infrastructure Engineer – Hewlett Packard Enterprise	2009 – 2010
Technical Associate – Minacs	2007 – 2009

DOCTORAL RESEARCH – MEDICAL AI SAFETY & INTERPRETABILITY

SAIL Lab (Secure and Assured Intelligent Learning), University of New Haven - Ph.D. Researcher 2021 - October 2026

- Dissertation: paraphrase sensitivity failures in medical vision-language models, where clinically equivalent questions produce contradictory diagnoses. Successfully defended August 26, 2026.
- Built PSF-Med, a benchmark of 92,856 audited question-paraphrase pairs from 26,850 chest X-ray questions across three countries; measured diagnosis flip rates of 8–58% across six medical VLMs.
- Applied sparse-autoencoder interpretability to locate causative features; mechanistically guided LoRA interventions reduced flip rates by 79.5% while preserving visual grounding.
- Released open benchmarks, models, and code on Hugging Face and GitHub (medical-vqa-robustness-analysis, psf-med).

EDUCATION & CERTIFICATIONS

Ph.D., Engineering and Applied Science (Data Science) – University of New Haven — Completed October 2026

M.S., Data Science – University of Connecticut School of Business (GPA 3.92) — 2017 – 2019

B.E., Electronics and Communication – Cochin University of Science and Technology — 2004 – 2008

Certifications: Tableau Desktop Specialist · Deep Learning Nanodegree, Udacity (2017) · Certificate in Project Management

SELECTED PROJECTS

- LIMRA Insurance Lapse Predictor: machine learning to understand and mitigate universal life insurance policy lapses among millennials.
- Precision oncology decision support: dynamic treatment planning for theranostics using pharmacokinetic modeling and reinforcement learning.

PUBLICATIONS

- Attention Without Grounding: Causal Evaluation of Visual Explanations in Medical VLMs. *iMIMIC Workshop, MICCAI, 2026.*
- Consistent but Dangerous: Per-Sample Safety Classification Reveals False Reliability in Medical Vision-Language Models. *Med-Reasoner Workshop, CVPR, 2026.*
- PSF-Med: Measuring and Explaining Paraphrase Sensitivity in Medical Vision-Language Models. *MMFM-BIOMED Workshop, CVPR, 2026.*
- Mechanistically Guided LoRA Improves Paraphrase Consistency in Medical Vision-Language Models. *CHIL, 2026.*
- Predictive Entropy Links Calibration and Paraphrase Sensitivity in Medical Vision-Language Models. *arXiv:2604.08941, 2026.*
- VSF-Med: A Vulnerability Scoring Framework for Medical Vision-Language Models. *IEEE ISBI, 2026.*
- When Chain-of-Thought Backfires: Evaluating Prompt Sensitivity in Medical Language Models. *Preprint, 2026.*
- Multimodal Deep Learning for Early Prediction of Patient Deterioration in the ICU. *Journal of Data Science, 2025.*
- Promise of Data-Driven Modeling and Decision Support for Precision Oncology and Theranostics. *arXiv:2505.09899, 2025.*
- Comparative Study of Generative Models for Early Detection of Failures in Medical Devices. *8th Intl. Conf. on Medical and Health Informatics, 2024.*
- Machine Learning for Anastomotic Leak Prediction: A Systematic Review. *Intl. J. of Trend in Scientific Research and Development, 2024.*

PATENTS

- System and method for machine learning to verify consistent staple line delivery. *WO2025027493A1, 2025.*
- Systems and methods for machine learning control of a surgical device. *WO2023223258A2, 2023.*

PROFESSIONAL SERVICE & PRESENTATIONS

- Peer reviewer: ICLR 2026, MICCAI 2026, Machine Learning for Healthcare (MLHC) 2026, IEEE ICMLA 2026.
- Lead Reviewer, International Medical Devices Safety Conference (2024, 2025).
- Posters: VSF-Med at IEEE ISBI 2026; PSF-Med at SMLM, Yale University, 2026.